Cambridge (CIE) A Level Computer Science

Your notes

Multimedia

Contents

- Bitmap Images
- Vector Graphics
- Sound Representation

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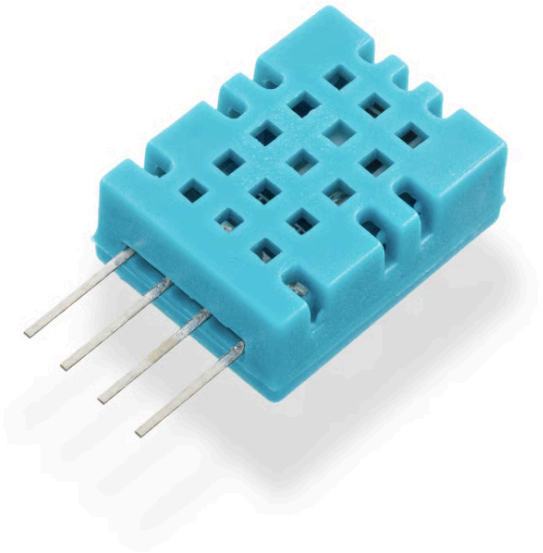
Your notes

Bitmap encoding

What is a bitmap?

- A bitmap image is made up of squares called **pixels**
- A pixel is the **smallest element** of a bitmap image
- Each pixel is **stored as a binary code**
- Binary codes are **unique** to the **colour** in each pixel
- A typical example of a bitmap image is a photograph



Humidity sensor bitmap image

- The more colours and more detail in the image, the **higher the quality of the image** and the **more binary that needs to be stored**

Image vs screen resolution

- **Image resolution** is the **total amount of pixels** that make up a bitmap image
- The image resolution is calculated by **multiplying the height and width** of the image (in pixels)
- In general, the **higher the resolution the more detail in the image** (higher quality)
- **Screen resolution** refers to the **total amount of pixels horizontally in a display**, such as:
  - **Computer monitors** - 1440p means 1440 pixels horizontally compared to 4K which is 3840 pixels (roughly 4 thousand)

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- **TVs** - HD (high definition) televisions have a screen resolution of 1080p, 1080 pixels horizontally compared to newer 4K televisions with 3840 pixels
- **YouTube** - The quality button allows a user to change the video playback resolution from 144p (144 pixels horizontally) up to 4K

- Another consideration of screen resolution is the **physical size** of the of the display
- The **number of pixels per square inch** (**PPI**) is known as **pixel density**
- Pixel density can mean images to **need to be scaled up or down to fit**, sometimes causing loss in quality

Case Study

- A consumer purchases a new **65" 4k television**
- The screen resolution is **3840 x 2160**
- To calculate the pixel density of the screen we **add together the squares of the resolution**
  - $(3840^2 + 2160^2) = (14\,745\,600 + 4\,665\,600) =$ **19 411 200**
  - Find the **square root** ($\sqrt{19\,411\,200}$ = **4405.814**)
  - **Divide by the screen size** (4405.814 / 65 = **68**)
- The television has a **pixel density of 68** pixels per inch (**PPI**)
- Watching **4K content** from a normal viewing distance means the image will appear **crisp and sharp**
- However, if you sit **too close**, you may start to see:
  - The **pixel grid**
  - A **loss of fine detail**
- Modern smartphones have **very high screen resolutions** in **much smaller screens**
- This gives them a much **higher PPI**, often over **300 PPI**
- This means they can be viewed **up close without losing quality** or seeing pixelation

Colour/bit depth

- Colour depth is the **number of bits stored per pixel** in a bitmap image
- The colour depth is dependent on the **number of colours needed** in the image
- In general, the **higher the colour depth the more detail in the image** (higher quality)
- In a black & white image the **colour depth would be 1**, meaning 1 bit is enough to create a unique binary code for each colour in the image (1=white, 0=black)

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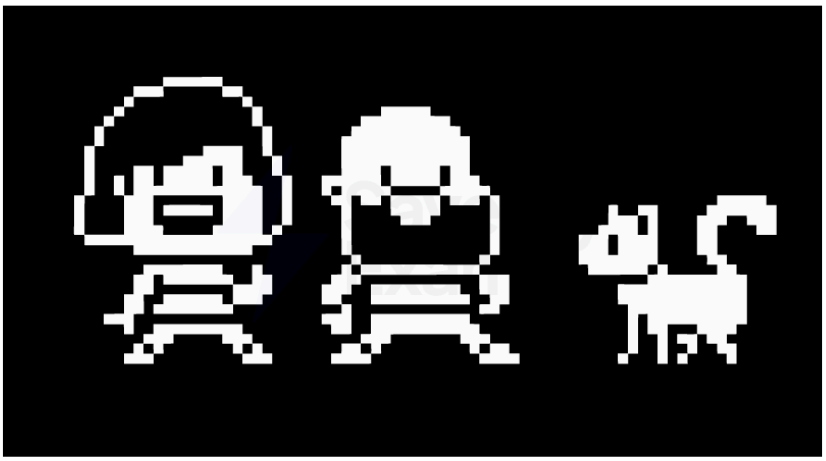

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- In an image with a **colour depth of 2**, you would have 00, 01, 10 & 11 available binary codes, so **4 colours**

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- As **colour depth increases**, so does the **amount of colours available** in an image
- **Colour depth** can also refer to the **number of colours** that can be represented in an image
- It is calculated using the formula:
  - **Colour depth = 2^n (where n = number of bits)**

Colour/bit depth	Number of colours	Example
1 bit	$2^1 = 1$	B&W
2 bit	$2^2 = 4$	Icons/logos
4 bit	$2^4 = 16$	Early computer graphics
8 bit	$2^8 = 256$	GIFs, retro games

24 bit (True colour)	2^{24} = 16 777 216	High-quality images

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Calculating the size of a bitmap file

How do you calculate the size of a bitmap image?

- Estimating the size of a bitmap image can be carried out with the following formula:
  - **Resolution x colour/bit depth**

Example

Image Files		
(Resolution) **x** (Colour Depth)		
Size of bitmap image =		
Resolution	**250,000**	Resolution = width x height
Colour depth	**24 bits**	24 bits = 3 bytes
250,000 x 24	**=** **(bit to bytes) /8** **(bytes to KiB) /1024** **or** **(bytes to KB) /1000**	**6,000,000 bits** **750,000 bytes** **732 KiB** **or** **750 KB**
250000 x 3	= **(bytes to KiB) /1024** **or** **(bytes to KB) /1000**	**750,000 bytes** **732 KiB** **or** **750 KB**

- When bitmap images are saved, **a file header is created**
- This contains:
  - **File type (.bmp or .jpg)**
  - **File size**
  - **Image resolution**

- **Colour depth**
- **Any type of compression if used**

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Worked Example

The following section of a bitmap image is 10 pixels wide and 5 pixels high. In this example, each colour is represented by a letter, e.g. O is orange.

The complete image can have up to 256 colours

B	B	B	B	B	B	B	B	B	B
Y	Y	P	Y	Y	Y	P	Y	Y	Y
R	R	M	R	P	K	T	T	R	R
B	O	P	Y	Y	Y	P	G	P	P
R	O	R	P	P	P	R	R	R	R

a) Identify the smallest number of bits that can be used to represent each colour in the complete bitmap image. **[1]**

b) Calculate an estimate for the file size of the section of the bitmap image shown, giving your answer in bytes. Use your answer from **part a** and show your working. **[2]**

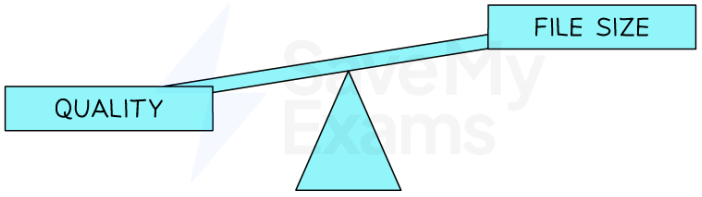
Answer

- 8 [1 mark]
- 10 x 5 = 50(pixels), 50 x 8(bits) = 400 // 400 ÷ 8(bits) = 50 [1 mark]
- 50 (bytes) [1 mark]

Image quality & file size

What is the relationship between image quality and file size?

- As the **resolution and/or colour depth increases**, the **bigger the size of the file** becomes on secondary storage
- The **higher the resolution**, the **more pixels** are in the image, the more bits are stored
- The **higher the colour depth**, the **more bits per pixel** are stored
- Striking a **balance between quality and file size** is always a consideration
- **Compression** can be used on high quality images to help reduce the file size



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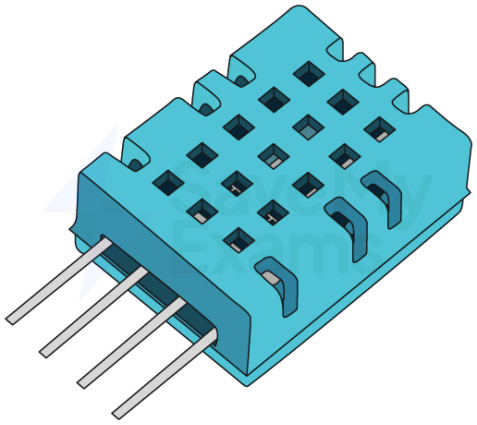
Vector graphics

What is a vector graphic?

- A vector graphic is created from **mathematical equations** and **points**
- Only the **mathematics used** to create the image are stored
- For example, to create a circle the data stored would be:
  - **Centre point (x, y coordinates)**
  - **Radius**
- Typical examples of vector images are **logos and clipart**



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- Vector images are infinitely **scalable**
- Ideal for situations where the same image **will be made bigger and smaller** and a **loss of quality is unacceptable**
  - For example, the same logo used on both a pencil and a billboard
- A vector graphic contains a **drawing list** (included in the file header)
- A typical drawing list will contain:
  - **Commands used to create each object in the image**
  - **Attributes that define the properties of each object**
  - **Relative position of each object**

- Dimensions of the image **are not defined**, meaning scaling up does not result in a loss of quality

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Bitmap vs vector

Feature	Bitmap Images	Vector Images
Made from	Pixels (tiny coloured squares)	Shapes and lines using maths (vectors)
File size	Larger (depends on resolution)	Smaller (uses formulas, not pixels)
Scalability	Loses quality when resized (pixelates)	Can be resized without losing quality
Best for	Photographs, detailed images	Logos, icons, text graphics
Editable with	Paint software (e.g. Photoshop, GIMP)	Drawing software (e.g. Illustrator, Inkscape)
Common file types	.jpg, .png, .bmp, .gif	.svg, .ai, .eps

- When **choosing** to use a bitmap or vector based image, the following should be considered:
  - **Does the image need to be resized?**
  - **Does the image need to be drawn to scale?**
  - **Does the image need to look real?**

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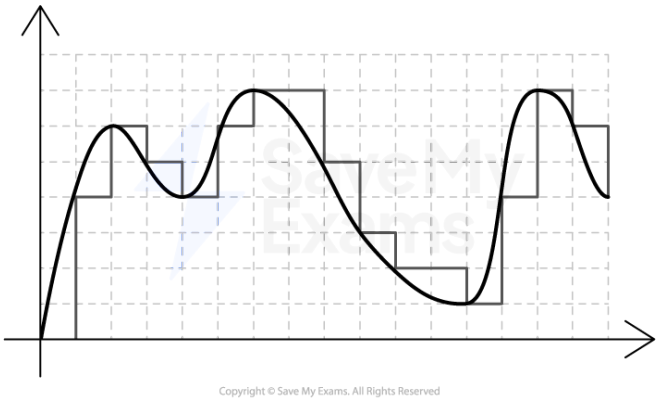
Sound encoding

How is sound encoded in a computer system?

- Computers represent all data in binary, including sound that we **record** using a microphone (input) or sound that we **playback** from a speaker (output)
- For this to happen, analogue sound must be **sampled and stored**

Analogue sound

- Sound waves begin as **analogue** and for a computer system to understand them they must be converted into a **digital form**
- Measurements of the original sound wave are **captured** and **stored as binary** on secondary storage
- This process is called **Analogue to Digital conversion** (**A2D**)
- The process begins by measuring the loudness (**amplitude**) of the analogue sound wave at a point in time, this is called **sampling**
- The higher the amplitude, the **louder** the sound
- Each measurement (**sample**) generates a value which can be represented in **binary** and **stored**
- Using the **samples**, a computer is able to create a **digital version** of the original analogue wave
- The digital wave is **stored** on **secondary storage** and can be played back at any time by reversing the process



- In this example, the grey line **represents the digital wave** that has been created by taking samples of the original analogue wave
- In order for the digital wave to **look more like the analogue wave** (black line) the **sampling rate**, **sampling resolution** and **sample interval** can be changed

Sampling rate vs sampling resolution

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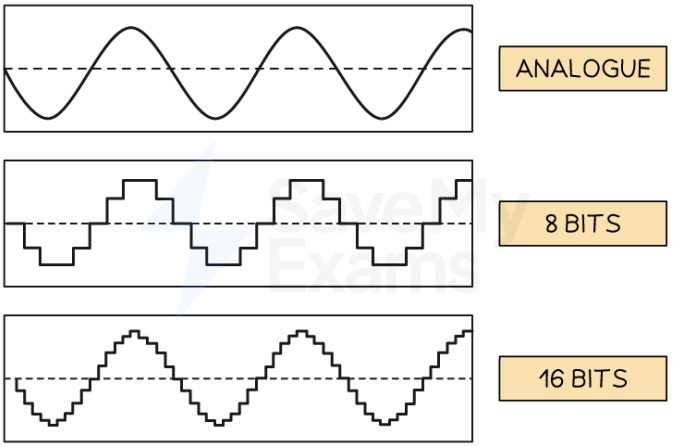
Sampling rate

- The sampling rate is the amount of **samples taken per second** of the analogue wave
- Samples are taken for the **duration** of the sound
- The sampling rate is measured in **Hertz** (**Hz**)
- **1 Hertz is equal to 1 sample** of the sound wave



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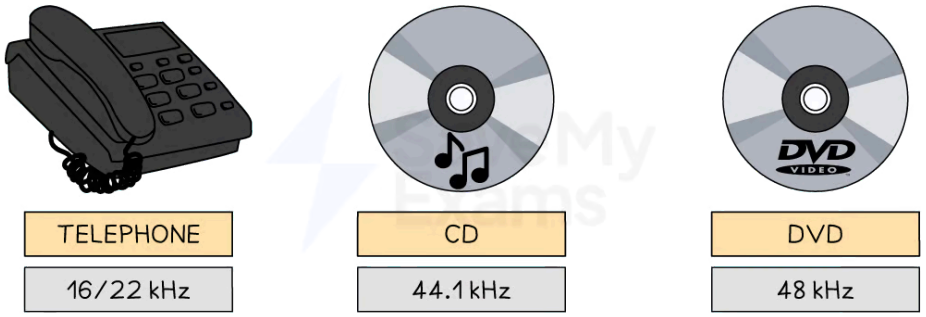
- In the example above, the higher the sample rate, the **closer to the original** sound wave the digital version looks

Sampling resolution

- Sampling resolution is the **number of bits** used to represent **each sound sample**
- Sampling resolution is closely related to the **bit depth of a bitmap image**, they measure the same thing in different contexts



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- The sample rate of a typical audio CD is **44.1kHz** (44,100 Hertz or 44,100 samples per second), a **sampling resolution of 16** and is recorded in **stereo sound**

- Using the graphic above to compare common sampling resolutions, the question, "*Why does telephone hold music sound so bad?*" can now be answered

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Impact of sampling settings

What are the impacts of sampling settings?

Factor	Effect of playback quality	Effect on file size
Sampling rate	↑**higher** = more detail, better sound quality	↑**higher** = more data, larger file size
Sampling resolution	↑**higher** = bigger range, better sound quality	↑**higher** = more data per sample, larger file size

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